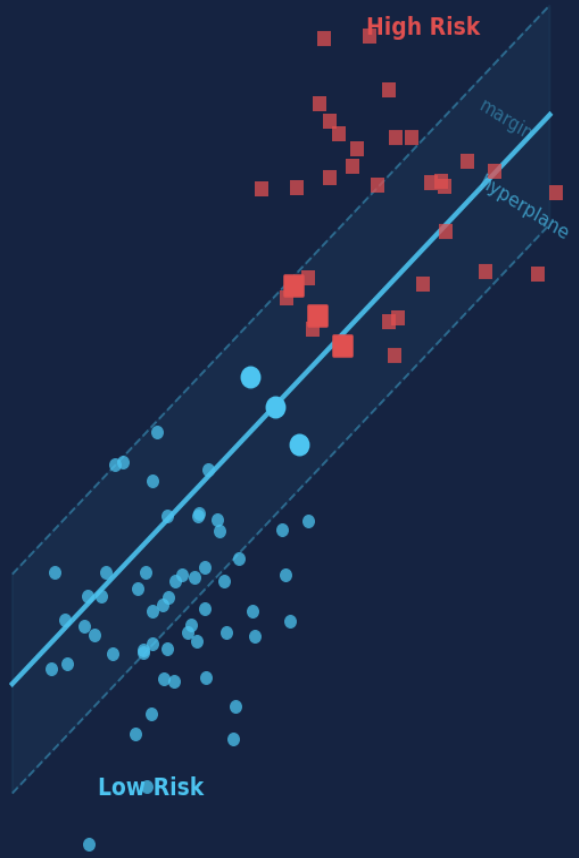
MACHINE LEARNING · HR ANALYTICS · PEOPLE RISK M

HR Employee Risk Analytics Are Not Random.

What if disengagement could be detected before it becomes attrition?

Support Vector Machine · Risk Scoring · Early Warning



1,000
Employee Records

10
Features

68.4%
Accuracy

0.659
ROC-AUC

The Real Problem

Disengagement, attrition, and performance decline rarely appear suddenly — they accumulate through measurable signals.

Attrition Cost

Replacing an employee costs 50–200% of their annual salary in recruiting, onboarding, training

Late Detection

HR typically identifies risk at resignation — after the signal has been visible for months

Ops Impact

High-risk employees correlate with scrap, quality failures and team performance drag

30.8% *of employees show behavioral patterns consistent with high risk — detectable months before any formal event.*

Every attrition event has a history in the data. The question is whether HR is reading it before the exit interview.

From Events to Probabilities

Old Thinking

Risk is reactive:

"The resignation happened."

Annual performance reviews.
Exit interviews after the fact.
No early signals. No prevention.
Just cost absorption.

New Thinking

Risk is probabilistic:

"How likely is this employee at risk?"

Score each employee quarterly.
Flag early signals before escalation.
Enable targeted HR conversations.
Turn behavioral data into decisions.

The right question is not 'Who left?' — it's 'Who is showing risk patterns right now, and what can we do today?'

The Data Behind the Problem

Three layers of employee data — behavioral, operational, and structural — each contributing a different dimension of risk:

BEHAVIORAL	**Punctuality rate, Engagement score** *Attendance patterns and self-reported engagement — the most direct disengagement signals*
OPERATIONAL	**Productivity index, Scrap %, Training hours, Experience** *Performance output and development investment — what the employee produces and receives*
STRUCTURAL	**Department, Shift, Contract type, Area rotation rate** *Organizational context — where the employee sits and under what conditions they operate*

Risk Rate by Structural Context

Contract	Temporary	40.8%	Permanent	25.7%	Outsourcing	34.2%
Department	Admin	38.9%	Production	37.4%	Quality	21.7%
Shift	Night	35.1%	Morning	31.3%	Afternoon	27.0%

Risk doesn't live in one variable — it emerges from specific combinations of behavior, performance, and context.

Why Support Vector Machine?

Employee risk doesn't follow a straight line. SVM finds the boundary that best separates risk patterns — even in non-linear feature space.

Non-linear Kernel (RBF)

Risk patterns combine multiple factors non-linearly. RBF kernel maps data into higher dimensions where a separating hyperplane exists — not possible with logistic regression.

Maximum Margin

SVM maximizes the margin between classes — it finds the boundary most likely to generalize. Especially valuable with limited labeled HR data.

Regularization via C

Parameter C balances fit vs. generalization. GridSearchCV found C=50, gamma=0.01 — tuned on F1 to prioritize recall on the minority risk class.

Linear Companion

RBF SVM predicts. LinearSVC explains. Same data, same preprocessing — the linear coefficients give HR the 'why' behind each risk flag.

Best kernel: RBF (C=50, gamma=0.01) — selected by GridSearchCV optimizing F1 over 5 stratified folds.

Model Results

68.4%	0.659	47.6%	26.0%
Accuracy	ROC-AUC	Precision (Risk)	Recall (Risk)

LinearSVC Coefficients — Risk Drivers

shift_Night	+0.268
scrap_associated_pct	+0.162
engagement_score	−0.234
contract_Permanent	−0.269
experience_yrs	−0.197
training_hours_annual	−0.181

Confusion Matrix (250 test employees)

	Pred: Low Risk	Pred: High Risk
Actual: Low	151 (TN)	22 (FP)
Actual: High	57 (FN)	20 (TP)

57 high-risk employees missed — each one an undetected intervention opportunity. Threshold tuning can improve recall.

Moderate performance is honest — HR data is noisier than sensor data. AUC > 0.5 confirms real signal exists.

From Insight to Action

Engagement −0.234	→ Strongest protective factor. Engagement survey scores below 2 trigger 1:1 HR conversation. This is the highest-leverage intervention available.
Permanent contract −0.269	→ Contract stability is protective — temporary and outsourced workers show 40.8% and 34.2% risk rates. Conversion to permanent is a structural lever.
Night shift +0.268	→ Night shift workers show 35.1% risk vs 27.0% afternoon. Consider rotation programs, shift incentives, and additional check-ins for night teams.
Scrap % +0.162	→ Scrap association signals operational stress or disengagement. Elevated scrap alongside low engagement is a compound flag — investigate both simultaneously.
Training hours −0.181	→ Investment in training is protective. Employees below 20 annual training hours are structurally underserved — prioritize them in development plans.

The model doesn't replace HR judgment — it tells them where to have the conversation, and when.

What-If Simulator — Employee Risk Scorecard

Same model. Three profiles. The third isolates the marginal effect of a targeted engagement intervention.

Employee A — Low Risk	
Training	50 hrs
Punctuality	99%
Productivity	110
Scrap	2.0%
Engagement	5/5
Contract	Permanent

28% risk probability

Employee B — High Risk	
Training	5 hrs
Punctuality	78%
Productivity	80
Scrap	10.0%
Engagement	1/5
Contract	Outsourcing

100% risk probability

Employee C — Intervention	
Training	5 hrs
Punctuality	78%
Productivity	80
Scrap	10.0%
Engagement	**4/5 ↑**
Contract	Outsourcing

99.6% risk probability

Scenario C insight: improving engagement alone (1→4) barely moves the needle from 100% to 99.6%. The risk pattern here is systemic — not a single-variable fix.

The score is a conversation starter — not a verdict. Context and human judgment determine the response.

This project is not about predicting who will leave.

It is about making disengagement visible while there's still time to act.

HR risk doesn't follow a linear rule. Risk emerges from combinations — and that's exactly why a kernel-based model captures it better.

Feel free to download the full project: dataset, notebook, and simulator.

Change the threshold. Adjust the kernel. Challenge what the model flags.

Because people development has no finish line.